UniFraud-GH: Semantic Concept Alignment and Prior-Aware Domain Adaptation for Cross-Sector Fraud Detection in Ghana

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***Abstract*—Financial fraud in Ghana spans payment cards, mobile money and national health insurance, but labels are plentiful in at most one sector at a time. Existing fraud-transfer work moves models between countries, banks or markets that share one schema; to our knowledge, and within the searches we ran, none transfers across sectors whose features and fraud priors both differ. We formalise this *heterogeneous cross-sector transfer* problem and make four contributions. (i) A *semantic concept alignment* maps each schema into a shared three-dimensional fraud-concept space, with proofs of unit-invariance and of exactly what the alignment can cost in Bayes risk. (ii) A unified treatment of adversarial, class-conditional kernel and fine-tuning adaptation, with a closed-form Jensen–Shannon result showing that exact marginal invariance forces an error floor under the sectors' priors (0.52% vs. 54.8% fraud), a prior correction that removes that obstruction under label shift, and the optimal weighting of scarce target labels. (iii) A leakage-free, label-budget-aware evaluation protocol with a transfer-gain metric and resampling-corrected significance tests. (iv) A formal audit of rule-generated labels, showing that scores on rule-labelled claims measure rule reproduction unless the rule inputs are excluded. The mathematical development is complete; empirical results are pending a corrected re-run.**

***Index Terms*—fraud detection, domain adaptation, transfer learning, label shift, class imbalance, maximum mean discrepancy, mobile money, health insurance, explainable AI.**

I. Introduction

FRAUD in Ghana's financial ecosystem is not confined to one channel. Card-not-present attacks, mobile-money social engineering and insurance-claim padding are carried out by overlapping populations of actors using recurring behavioural tactics: abnormal amounts, exploitation of weakly-vetted counterparties, and structural irregularities in time, geography or billing. Labelled examples are nevertheless siloed and uneven: card fraud is abundant but extremely rare per transaction, mobile-money simulators produce balanced traffic, and insurance claims are few and labelled mainly by tariff rules [1], [2].

This situation calls for transfer across *heterogeneous* domains: different feature spaces $\mathcal{X}_k$ *and* different label priors π_k. Two facts make it hard. First, conventional domain adaptation presupposes a common input space. Second, even with a common representation, exact invariance has a provable error floor, and can hurt, when the label marginals differ [3]. Existing fraud-transfer studies sidestep both difficulties by transferring between countries, markets or banks that share a schema [4]–[8]. The cross-sector problem is therefore not an incremental variation but a different statistical setting, which we study from first principles and keep tied to claims that can be proved or tested.

Manuscript draft November 3, 2026. Numerical results in Section IX are placeholders (**TBD**) pending re-execution of the corrected, leakage-free protocol of Section VIII. Reference details were located through search records and must be checked against publisher records before submission.

Contributions and novelty

1) **Problem and positioning.** We define heterogeneous cross-sector fraud transfer (Definition 1) and show, through a systematic comparison (Table II), that the surveyed fraud-transfer literature assumes a shared schema and a single sector. This is the *cross-domain application* novelty: card, mobile-money and health-insurance fraud are connected through a common behavioural vocabulary rather than a common table.
2) **Semantic concept alignment** (Section V-B). A schema-to-concept operator $\phi_k : \mathbb{R}^{d_k} \to \mathbb{R}^3$ with proofs of unit-invariance and orientation consistency (Prop. 3) and a Bayes-risk monotonicity result (Prop. 4) that states exactly when alignment is lossless.
3) **Prior-aware adaptation theory** (Sections V-E, VII). A unified objective for DANN, class-conditional MMD and fine-tuning; the saddle-point characterisation of gradient reversal (Prop. 7); a closed-form Jensen–Shannon lower bound evaluated on the actual priors of the three sectors (Prop. 9, Table VI); a label-shift correction and a black-box prior estimator (Props. 10, 11); the optimal labelled-target weighting (Prop. 12); and a bandwidth analysis of the MMD criterion (Prop. 13).
4) **Prior-aware decision theory and operating points** (Sections V-C, VII-E). Exact relations among weighted loss, resampling and the Bayes-optimal threshold (Props. 5, 6), and the sector-specific false-positive budgets implied by prevalence (Prop. 14), which explain why a single metric is not comparable across sectors.
5) **Leakage-free protocol and label audit** (Sections IV, VIII). A formal account of rule-label circularity and of a missing-data artefact in the health-insurance rules (Props. 2, 1), with a transfer-gain metric that isolates transfer from supervision.

II. Related Work

We organise the literature by the question each strand answers, and close with a positioning that states what is, and

is not, new. All claims of absence are qualified: they rest on the searches we ran and not on an exhaustive survey.

A. Fraud detection in the three sectors

Credit cards. Practical card-fraud systems face concept drift, extreme imbalance and *verification latency*: investigators label only a few alerts quickly, and the remaining labels arrive late. Dal Pozzolo et al. [9] formalise this setting and train separate classifiers on feedback and on delayed samples, validating on over 75 million real transactions; Carcillo et al. [10] describe a streaming Spark architecture for the same regime. Both are card-specific, proprietary, and say nothing about portability to other products. Graph-based methods add relational structure within a sector [11], but a card-merchant graph has no direct analogue in mobile money or claims, and on a Bitcoin benchmark random forests outperformed graph convolutional networks [12], cautioning against assuming that structure beats strong tabular baselines. Unsupervised detectors such as isolation forests [13] need no labels but assume that fraud is rare *and* different, and their scores are not comparable across domains.

Mobile money. Public mobile-money data are scarce because transactions are sensitive, so the field relies on agent-based simulators: PaySim, calibrated on logs from an African mobile-money service [14], and MoMTSim, which adds Sub-Saharan fraud scenarios such as SIM swap and account takeover [2], [15], [16]. Comparative studies of SVMs, boosted trees and naive Bayes on mobile-money fraud [17] and of logistic regression, random forests and XGBoost on MoMTSim [16] are single-sector and, for the latter, entirely simulated. Dataset notes for PaySim warn that balance columns leak the label, and MoMTSim fidelity is judged on marginal statistics, which says little about whether fraud *structure* transfers.

Health insurance. The Ghanaian National Health Insurance claims literature is thin. Sowah et al. [1] use a genetic-algorithm-tuned SVM on real claims and report accuracy of 80–88%; accuracy is a weak headline metric under imbalance, and we found no independent replication, temporal evaluation, or link to other sectors. International medical-claims fraud is outside our verified searches.

B. Synthetic data and realistic evaluation

Because two of our three sectors are simulated, evaluation hygiene is central. Altman et al. [18] show how a transaction generator's assumptions should be documented, and Jordon et al. [19] survey what synthetic data can and cannot support; Grover et al. [20] and Jesus et al. [21] provide public fraud benchmarks with standardised splits, drift and injected bias, though each is single-domain. Data leakage—preprocessing, encoding or selection fitted before splitting, or features derived from labelling rules—is a documented cause of irreproducible results across fields [22], [23], and high-cardinality target encoding is a classic channel [24]. Resampling distorts posterior probabilities [25], [26], so thresholds must be set on unresampled validation data. For imbalanced metrics, precision–recall analysis is more informative than ROC [27] and MCC is more robust than F_1 or accuracy [28]; both depend on prevalence, which differs by two orders of magnitude across sectors (Prop. 14). Statistical comparison requires corrections for the dependence of resampled estimates [29], non-parametric or Bayesian alternatives [30], [31], and multiplicity control [32].

C. Transfer learning in fraud detection

Transfer in fraud has been studied where source and target share a feature schema. Lebichot et al. [4] compare 15 transfer strategies across European countries on over 200 million e-commerce card transactions and propose a semi-supervised domain-adaptation ensemble that is largely insensitive to the number of target labels; Siblini et al. [5] report the same programme taken to production. Zhu et al. [6] propose a Hierarchical Explainable Network (HEN) over user behaviour sequences with a transfer framework from mature to new cross-border markets; "domain" there means country or market within one platform and sequence format. Pereira et al. [7] evaluate transfer methods on bank-account-fraud streams under drift, again with a shared schema, and federated meta-learning lets banks with a common feature space collaborate without pooling data [8]. Transfer *between* fraud sectors with different schemas was not found in these searches; Section II-H states the resulting claim carefully. The terminology also needs care: our attention-gated network is an *adaptation* of the HEN idea of [6], and it is not their hierarchical sequence model.

D. Domain-adaptation theory and methods

The foundational bound of Ben-David et al. [33] limits target error by source error, the $\mathcal{H}\Delta\mathcal{H}$-divergence between marginals, and a joint-optimal error $\lambda^\star$ (Eq. (1)); Mansour et al. [34] generalise to arbitrary losses through the discrepancy distance, which is estimable from unlabelled data. Practical methods fall into adversarial and kernel families. DANN [35] trains features to confuse a domain discriminator through gradient reversal, ADDA [36] separates source and target encoders, and CDAN [37] conditions the discriminator on classifier predictions to align joint structure. Kernel methods match moments: DAN [38] minimises multi-kernel MMD [39] across layers, CORAL [40] matches second-order statistics, optimal transport [41] learns a mass-preserving map (problematic when class masses differ), and subdomain methods [42] match class-conditional distributions with local MMD. Surveys are in [43].

E. Label shift, heterogeneity and negative transfer

Cross-sector fraud transfer is dominated by two issues that most methods above do not treat. *Label shift*: Zhao et al. [3] prove an information-theoretic lower bound showing that exact marginal invariance *forces* error when label marginals differ, motivating methods that estimate and correct the prior [44], [45], that handle target and conditional shift jointly through kernel embeddings [46], or that match conditionals under *generalised* label shift [47]; Johansson et al. [48] add that the aligning map must remain invertible on the support. *Heterogeneity*: shared-space methods require a common input

dimension; heterogeneous feature augmentation [49] learns projections into a common subspace but needs target labels and offers no guarantee that the subspace is semantically shared. Schema-matching benchmarks [50] match columns by name and value overlap, not by shared latent concepts. *Negative transfer*: transfer can hurt, and a no-transfer baseline must be reported [51]; TableShift [52] finds that tabular distribution-shift gaps are strongly associated with label shift and that gradient-boosted trees remain strong baselines. Model selection without target labels is unreliable under large prior mismatch [53], and regularised fine-tuning toward pre-trained weights [54] is a sensible few-label default.

F. Tabular learners, explanation and weak labels

Gradient-boosted trees [55] and random forests [56] remain the default for tabular data [57], [58], with competitive deep alternatives [59] and, for small samples such as our 2 655 claims, tabular foundation models [60]; stacked generalisation [61] combines them. Explanations are expected in financial decisions [62]: SHAP [63], [64] and LIME [65] are standard, but Shapley importances have known conceptual limits [66] and perturbation-based explainers can be fooled off the data manifold [67], which matters when aligned features are off-manifold. Fairness surveys [68] note that protected attributes may be present in one schema and absent in another. Fraud rules are natural *labelling functions* for weak supervision [69], [70], and noisy-label learning [71] applies, but fraud label noise is class-dependent (missed fraud is labelled legitimate) and, in our rule-labelled claims, driven by missing data (Prop. 1).

G. Critical synthesis of the most relevant studies

Table I condenses what each closely related study establishes and what it cannot tell us about cross-sector transfer. Three patterns emerge. First, *every fraud-transfer result we found fixes the schema*: gains from adaptation [4]–[8] come from correcting covariate or temporal shift inside one product line, so they do not bear on whether behavioural knowledge survives a change of sector. Second, *the theory that does cover our setting is negative*: with differing priors and non-shared conditionals, bounds are vacuous unless the joint error $\lambda^\star$ is small [3], [33], [34], and heterogeneous-feature methods lack semantic guarantees [49]. Third, *the evaluation practices that would expose this are rarely applied*: target-only baselines and negative-transfer gaps [51], prevalence-aware metrics [27], and leakage controls [22] are absent from most applied fraud studies, and benchmark shift studies find that label shift dominates the gap [52]. These three patterns define the design requirements of Section V.

H. Positioning and novelty

Table II summarises the comparison. *To our knowledge, and within the searches we ran*, the new elements are: (a) the **cross-sector setting** itself—transfer among card, mobile-money and health-insurance fraud with different schemas; (b) **concept-space alignment** through a documented, orientation-aware operator that connects heterogeneous schemas by shared fraud concepts rather than by columns [50] or learned projections [49]; (c) the **prior-shift analysis** that quantifies, from the actual sector prevalences, in which transfer directions exact invariance carries an error floor (and in which it is benign), with the corrective machinery; (d) a **label audit and leakage-free protocol** for rule-labelled claims; and (e) a **transfer-gain metric** defined against a target-only baseline trained on the same labels. The individual components (DANN, MMD, fine-tuning, SHAP) are standard; the contribution is their principled composition for a setting where the standard assumptions fail, together with a proof-backed account of *why* and *where*. We do not claim priority beyond the evidence above; a Scopus/IEEE Xplore/ACM DL search should be rerun before submission.

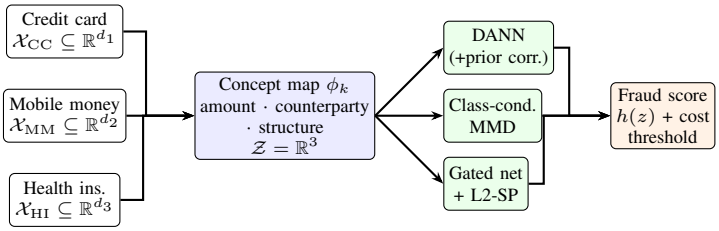


Fig. 1. UniFraud-GH pipeline: sector schemas are projected into a shared concept space, three adaptation strategies operate there, and the resulting score is thresholded with a cost-sensitive rule.

III. Problem Formulation

Let $k \in \mathcal{K} = \{\mathrm{CC}, \mathrm{MM}, \mathrm{HI}\}$ index sectors. Sector k has a feature space $\mathcal{X}_k \subseteq \mathbb{R}^{d_k}$, label space $\mathcal{Y} = \{0, 1\}$ (1 = fraud), and joint distribution $\mathcal{D}_k$ over $\mathcal{X}_k \times \mathcal{Y}$. A labelled sample $\mathcal{S}_k = \{(x_i, y_i)\}_{i=1}^{n_k} \sim \mathcal{D}_k^{n_k}$ is available, with prior $\pi_k = \mathbb{P}_{\mathcal{D}_k}(Y = 1)$ (notation in Table III).

Definition 1 (Heterogeneous cross-sector transfer)**.** *Given a source sector s and a target sector $t \neq s$ with $\mathcal{X}_s \neq \mathcal{X}_t$ and $\pi_s \neq \pi_t$, together with a* label budget *$b \in \{0\} \cup \mathbb{N}$ of target labels ($b = 0$: unsupervised adaptation), find a predictor $h : \mathcal{X}_t \to [0, 1]$ minimising the target risk $R_t(h) = \mathbb{E}_{(x,y)\sim\mathcal{D}_t}\, \ell(h(x), y)$ using $\mathcal{S}_s$, the unlabelled target features, and at most b target labels.*

For a hypothesis class $\mathcal{H}$ over a common space $\mathcal{Z}$, write $\varepsilon_k(h) = \mathbb{P}_{\mathcal{D}_k}(h(z) \neq y)$. The central bound [33] is

$$\varepsilon_t(h) \;\leq\; \varepsilon_s(h) + \tfrac{1}{2}\, d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{D}_s^Z, \mathcal{D}_t^Z) + \lambda^\star, \qquad (1)$$

where $\lambda^\star = \min_{h' \in \mathcal{H}} [\varepsilon_s(h') + \varepsilon_t(h')]$. Eq. (1) is informative only if (a) a common space $\mathcal{Z}$ exists (Section V-B) and (b) $\lambda^\star$ is small. Condition (b) is the crux for fraud: no unlabelled alignment can reduce $\lambda^\star$, and under label shift it is bounded away from zero (Prop. 9).

IV. Data and Label Construction

Sparkov and MoMTSim are synthetic generators [2], [72]; external validity to production data is therefore a stated limitation (Section X), and generator assumptions should be documented as recommended in [18]. The health-insurance claims are real but carry no adjudicated fraud flag; labels are

TABLE I
WHAT CLOSELY RELATED STUDIES ESTABLISH, AND WHAT THEY LEAVE OPEN FOR CROSS-SECTOR FRAUD TRANSFER.

Study	Establishes	Limitation relative to our setting
Dal Pozzolo et al. [9]	Realistic card-FDS model: feedback vs. delayed labels, drift, imbalance; real data at scale	Card-only, proprietary; transfer not addressed
Lebichot et al. [4]	Semi-supervised DA across countries is competitive and insensitive to number of target labels	Same schema and sector; no prior-shift analysis
Siblini et al. [5]	Research-to-production path for cross-country card transfer, $\sim$15% gain over zero-shot	Homogeneous features; proprietary
Zhu et al. [6]	Hierarchical explainable network and transfer across markets of one platform	"Domain" = market; behaviour sequences, not tabular cross-sector
Pereira et al. [7]	Framework to evaluate transfer under drift with scarce target labels	Shared bank-account schema
Zheng et al. [8]	Federated meta-learning without pooling bank data	Common feature space
Botchey et al. [17]; Azamuke et al. [16]	Classical ML comparisons for mobile-money fraud, simulated data (MoMTSim)	Single sector; generator-defined fraud
Sowah et al. [1]	Real Ghana NHIS claims fraud model (GSVM), accuracy 80–88%	Accuracy metric, single sector, no replication
Zhao et al. [3]	Lower bound: invariance + label shift $\Rightarrow$ error	Not applied to fraud priors before this work
Tachet des Combes et al. [47]	Conditional matching + class reweighting under generalised label shift	Needs target class weights; heterogeneity not addressed
Duan et al. [49]	Projections for heterogeneous feature spaces	Needs target labels; no semantic-sharing guarantee; no prior shift
Gardner et al. [52]	Tabular shift benchmark: label shift drives gap; GBDTs strong	Shared schema; no fraud sectors

TABLE II
POSITIONING AGAINST THE CLOSEST RELATED WORK. "SCHEMA" = WHETHER SOURCE AND TARGET SHARE A FEATURE SCHEMA. "PRIOR SHIFT" = WHETHER DIFFERING FRAUD PRIORS ARE ANALYSED.

Work	Sector(s)	Setting	Schema	Prior shift	Data
Dal Pozzolo et al. [9]	card	drift, label latency	single	no	real, proprietary
Lebichot et al. [4]	card	cross-country DA	shared	no	real, proprietary
Siblini et al. [5]	card	cross-country, production	shared	no	real, proprietary
Zhu et al. (HEN) [6]	e-commerce card	cross-market, sequences	shared	no	real
Pereira et al. [7]	bank accounts	streaming transfer	shared	no	real + public BAF
Zheng et al. [8]	card	federated meta-learning	shared	no	real
Azamuke et al. [16]	mobile money	single-sector ML	single	no	simulated
Botchey et al. [17]	mobile money	model comparison	single	no	see paper
Sowah et al. [1]	health insurance	GSVM	single	no	real, Ghana NHIS
Duan et al. [49] (general)	—	heterogeneous DA	different	no	vision/text
This work	**card, mobile money, health ins.**	**cross-sector transfer**	**different**	**yes (analysed, corrected)**	**2 simulated + 1 real**

TABLE III
NOTATION.

Symbol	Meaning
$k \in \{\mathrm{CC}, \mathrm{MM}, \mathrm{HI}\}$	sector (card, mobile money, health ins.)
$\mathcal{X}_k \subseteq \mathbb{R}^{d_k}$, $\mathcal{D}_k$	feature space and joint law of sector k
$\pi_k = \mathbb{P}_{\mathcal{D}_k}(Y{=}1)$	fraud prior
$\phi_k : \mathbb{R}^{d_k} \to \mathbb{R}^3$	concept map (Eq. (4))
G_f, G_y, G_d	feature extractor, label head, domain head
$\varepsilon_k(h)$	0–1 error of h under $\mathcal{D}_k$
$d_{\mathcal{H}\Delta\mathcal{H}}$	$\mathcal{H}\Delta\mathcal{H}$-divergence
$\lambda^\star$	joint-optimal error (Eq. (1))
ω	positive-class weight; q = weighted posterior
b	number of labelled target examples
JS, d_{JS}	Jensen–Shannon divergence ($\log_2$), distance
$\widehat{\mathrm{MMD}}^2$	biased squared MMD estimator
TPR, FPR	true / false positive rate

TABLE IV
DATASET STATISTICS (VERIFIED AGAINST NOTEBOOK OUTPUTS).

Sector	n_k	π_k	Label provenance
Credit card (Sparkov)	1 852 394	0.521%	simulator ground truth
Mobile money (MoMTSim)	550 358	54.84%	simulator ground truth
Health ins. (NHIA)	2 655	41.62%	*rule-generated*

produced by tariff-compliance rules attributed in the project repository to [1] (to be verified).

A. Rule-generated labels

Let a claim have service charge S, medical (drug) charge M, total T and approved tariff A, the latter obtained by joining the claim's G-DRG code to the tariff table (defined only when the join succeeds; match rate 67.1%, 1 781 of 2 655). With $\mathbf{1}[\cdot]$ the indicator and $\mathrm{na}(u) = 0$ for missing u, a direct implementation of the tariff rules, in which missing charges are treated as zero,

yields

$$r_1 = \mathbf{1}[S > A], \quad r_2 = \mathbf{1}[M > \tfrac{1}{2}A], \quad r_3 = \mathbf{1}[T > A],$$
$$r_4 = \mathbf{1}[\,|T - (S+M)| > 1\,], \quad r_5 = \mathbf{1}[\min\{\mathrm{na}(S), \mathrm{na}(M), \mathrm{na}(T)\} \le 0],$$
$$y = \mathbf{1}\Big[\textstyle\sum_{i=1}^{5} r_i > 0\Big]. \tag{2}$$

Observed trigger counts are $(8, 271, 470, 6, 632)$ for $(r_1, \ldots, r_5)$.

Proposition 1 (Missing-data artefact in r_5)**.** *Let $n = 2655$, $n_M = 629$ the number of claims with M missing (since M is observed for 2026 claims), and $n_5 = 632$ the number of claims on which r_5 fires. Then at least $n_M/n_5 \approx 99.5\%$ of r_5's triggers are explained by a missing drug charge, and r_5 alone accounts for at most $n_5/\sum y = 632/1105 \approx 57.2\%$ of positive labels. The implied fraud prior excluding r_5 is at least $(1105 - 632)/2655 \approx 17.8\%$ and at most 41.62%; hence the reported prevalence conflates fraud with missingness.*

Proof. Under r_5, $\mathrm{na}(M) = 0 \le 0$ holds whenever M is missing, so every missing-M claim triggers r_5; at most $n_5 - n_M = 3$ triggers can come from other causes. The remaining counts follow by subtraction, with equality only if no other rule fires on r_5-flagged claims. □

Missingness is thus being labelled as fraud, a defect of this naive construction and not a property of fraud; the corrected pipeline treats missing data as a data-quality flag and reports results with and without r_5's intent (observed non-positive bills) and with unmatched claims excluded. Because missingness of M is plausibly informative of facility or claim type (not necessarily fraud), a classifier given the missingness pattern will exploit it, which is a form of weak-label noise [69], [71] correlated with covariates and therefore harder to remove than uniform noise.

Proposition 2 (Circularity of rule-generated labels)**.** *Suppose $y = g(x_R)$ for a deterministic g and a feature subset R. If a model class contains a function of x_R that equals g (e.g. a depth-bounded tree for threshold rules), then the Bayes risk over the full feature set is 0, and any held-out score on that feature set estimates the* rule-reproduction *ability of the learner rather than fraud detection. Moreover, any feature $f(x_R)$ that is a function of rule inputs inherits this leakage.*

Proof. $h(x) = g(x_R)$ attains zero error under $\mathcal{D}_t$ by construction, so $\min_h R_t(h) = 0$. A held-out estimate of $R_t(\hat{h})$ then measures how closely $\hat{h}$ approximates g. For $f(x_R)$: g is a function of x_R, hence also of $(x_R, f(x_R))$. □

In particular the features S, M, T, A, ratios and deviations $(\cdot - A)/A$, the per-DRG label rate, and quasi-identifiers such as folder and insurance-ID numbers (which allow the same patient to appear in train and test) must be excluded, and splits must be grouped by patient. These are the leakage channels catalogued generally in [22], [23].

V. Methodology

A. Leakage-free preprocessing

Let $\mathcal{I}_{\mathrm{tr}}, \mathcal{I}_{\mathrm{va}}, \mathcal{I}_{\mathrm{te}}$ be a partition of indices (random stratified for HI grouped by patient; *chronological* for CC and MM). Every data-dependent transform T_θ (imputation, capping at the q-th percentile, scaling, categorical encoding, feature selection) is fit on $\mathcal{I}_{\mathrm{tr}}$ only: $\hat{\theta} = \theta(\{x_i\}_{i \in \mathcal{I}_{\mathrm{tr}}})$, then applied unchanged to validation and test. For a categorical value c with n_c training rows, mean label $\bar{y}_c$ and overall mean $\bar{y}$, the smoothed target encoding is

$$\mathrm{TE}(c) = \frac{n_c \bar{y}_c + m\,\bar{y}}{n_c + m}, \tag{3}$$

[24], computed from *out-of-fold* statistics within the training partition so that a row's own label never enters its encoding. Resampling (below) is applied inside each training fold only.

B. Semantic concept alignment

Let $F_k = \{1, \ldots, d_k\}$ index sector-k features. A domain expert (or a documented rule table) provides, for each concept $c \in \{\text{amt, cpty, dist}\}$, a disjoint index set $G_{kc} \subseteq F_k$ and an orientation $\sigma_j \in \{-1, +1\}$ for $j \in G_{kc}$ ($+1$ iff larger values indicate higher risk).

Definition 2 (Concept map)**.** *Let $\mu_j, s_j > 0$ be the training mean and standard deviation of feature j. The concept map $\phi_k : \mathbb{R}^{d_k} \to \mathbb{R}^3$ has components*

$$\phi_k(x)_c = \frac{1}{|G_{kc}|} \sum_{j \in G_{kc}} \sigma_j \, \frac{x_j - \mu_j}{s_j}, \qquad \phi_k(x)_c := 0 \text{ if } G_{kc} = \emptyset. \tag{4}$$

Proposition 3 (Affine invariance and orientation consistency)**.** *(i) If feature j is replaced by $a_j x_j + b_j$ with $a_j > 0$ (a change of units, e.g. USD→GHS), then $\phi_k(x)_c$ is unchanged. (ii) Each component has training mean 0, and, if features in G_{kc} have pairwise correlations ρ_{jl} (after orientation), $\mathrm{Var}(\phi_k(x)_c) = |G_{kc}|^{-2} \sum_{j,l \in G_{kc}} \rho_{jl}$, which lies in $[\bar{\rho}, 1]$ with $\bar{\rho}$ the mean pairwise correlation. (iii) If all oriented features in G_{kc} are monotone non-decreasing in a latent risk ρ, then $\phi_k(x)_c$ is monotone non-decreasing in ρ.*

Proof. (i) With $\mu'_j = a_j \mu_j + b_j$, $s'_j = a_j s_j$: $(a_j x_j + b_j - \mu'_j)/s'_j = (x_j - \mu_j)/s_j$. (ii) Linearity of expectation gives mean 0; the variance is the average of all $|G|^2$ entries of the correlation matrix, whose diagonal contributes $|G|^{-1}$ and whose off-diagonal contributes $(1 - |G|^{-1})\bar{\rho}$, hence $\mathrm{Var} = |G|^{-1} + (1 - |G|^{-1})\bar{\rho} \in [\bar{\rho}, 1]$. (iii) A non-negative combination of non-decreasing functions is non-decreasing. □

Alignment is useful only if it does not discard fraud-relevant information. The next result states the price exactly.

Proposition 4 (Bayes-risk monotonicity of alignment)**.** *Let $\eta(x) = \mathbb{P}(Y{=}1 \mid X{=}x)$, $R^*(X) = \mathbb{E}[\min\{\eta(X), 1 - \eta(X)\}]$ the Bayes error in the full space, and $R^*(\phi(X))$ that in concept space. Then $R^*(\phi(X)) \ge R^*(X)$. Equality holds iff, almost surely, conditionally on $\phi(X)$ the quantity $\eta(X) - \frac{1}{2}$ has a constant sign, i.e. iff some Bayes-optimal classifier on $\mathcal{X}$ is a function of $\phi(X)$. In particular equality holds whenever ϕ is sufficient for Y (i.e. η is a function of $\phi(X)$), but sufficiency is not necessary.*

Proof. Write $\min\{u, 1-u\} = \frac{1}{2} - |u - \frac{1}{2}|$. With $\eta_\phi = \mathbb{E}[\eta(X) \mid \phi(X)]$ we have $R^*(X) = \frac{1}{2} - \mathbb{E}|\eta - \frac{1}{2}|$ and $R^*(\phi(X)) = \frac{1}{2} - \mathbb{E}|\eta_\phi - \frac{1}{2}|$ (the posterior given $\phi(X)$ is η_ϕ). Hence $R^*(\phi(X)) -$

$R^*(X) = \mathbb{E}\big[\mathbb{E}\big(|\eta - \frac{1}{2}| \mid \phi\big) - \big|\mathbb{E}(\eta - \frac{1}{2} \mid \phi)\big|\big] \geq 0$ by the conditional Jensen (triangle) inequality $\mathbb{E}|Z| \geq |\mathbb{E}Z|$. Equality in $\mathbb{E}[|Z| \mid \phi] \geq |\mathbb{E}[Z \mid \phi]|$ holds a.s. iff $Z = \eta - \frac{1}{2}$ is a.s. non-negative or a.s. non-positive given ϕ, which is exactly the stated condition. □

Thus within-domain models in the full space are never worse than concept-space models in the Bayes sense; the concept map is justified solely by *transfer*, where the full spaces are not comparable. This is why we report, for each sector, the concept-space score alongside the full-feature score.

Remark 1. *Sign orientation σ_j is essential: without it, averaging a risk-increasing and a risk-decreasing feature cancels the signal. Pairing raw columns across sectors by position (the first $\min(d_s, d_t)$ columns of each) would be semantically meaningless, because positional columns have unrelated meanings, and it would break the premise of* (1)*; we include it only as a negative-control baseline (A1), whose behaviour we expect to be order-dependent and unstable (it changes with column order) rather than uniformly worse. The shared space is therefore $\mathcal{Z} = \mathbb{R}^3$ (optionally augmented with a shared set of engineered universal features such as the robust amount z-score and a velocity proxy).*

C. Class imbalance and decision rule

Let $p(x) = \mathbb{P}(Y = 1 \mid x)$ under the deployment prior π.

a) Weighted loss.: With positive-class weight $\omega > 0$ the empirical risk is $\hat{R}_\omega(f) = \frac{1}{n}\sum_i[\omega y_i \ell(f(x_i), 1) + (1 - y_i)\ell(f(x_i), 0)]$ where ℓ is the logistic loss.

Proposition 5 (Weighting is a prior shift)**.** *The population minimiser of R_ω over all measurable f yields probabilities $q(x) = \sigma(f^\star(x))$ with*

$$\frac{q(x)}{1 - q(x)} = \omega \, \frac{p(x)}{1 - p(x)}. \tag{5}$$

Consequently $p(x) = q(x)/\big(q(x) + \omega(1 - q(x))\big)$. The same relation holds, with $\omega = \frac{\pi'/(1-\pi')}{\pi/(1-\pi)}$, for training on data resampled by replication to prior π' (and approximately for SMOTE, whose synthetic points leave the class-conditional density only approximately unchanged [25], [26]).

Proof. Pointwise, the minimiser of $p\,\omega\,\ell(f, 1) + (1-p)\,\ell(f, 0)$ is the logistic-calibrated value for the normalised weights $\tilde{p} = \omega p/(\omega p + 1 - p)$, i.e. $\sigma(f^\star) = \tilde{p}$. Odds scale by ω. □

b) SMOTE.: For a minority point x_i with a k-nearest minority neighbour x_ν, SMOTE generates $x' = x_i + \gamma(x_\nu - x_i)$, $\gamma \sim \mathrm{U}[0, 1]$ [25]. Synthetic points lie in the convex hull of minority neighbours; applied before data splitting or inside cross-validation *outside* the fold loop it places near-copies of validation points in training folds, inflating scores. We apply it inside each training fold and compare it with reweighting (XGBoost's per-instance weights [55]) as an ablation.

Proposition 6 (Cost-sensitive Bayes threshold)**.** *With costs $c_{FP}, c_{FN} > 0$ for false positives and false negatives (zero for correct decisions), the decision minimising expected cost predicts fraud iff $p(x) \geq \tau^\star = c_{FP}/(c_{FP} + c_{FN})$. In terms of a model trained with weight ω, the equivalent threshold on q is $\tau_q = \omega\tau^\star/(\omega\tau^\star + 1 - \tau^\star)$.*

Proof. Predicting 1 costs $(1-p)c_{FP}$, predicting 0 costs $p\,c_{FN}$; the first is no larger iff $p \geq c_{FP}/(c_{FP} + c_{FN})$. Substituting Prop. 5 gives τ_q. □

Operationally, thresholds are chosen on a *validation* partition (never on test) either by maximising F_1/MCC or, preferably, by Prop. 6 with sector-specific costs.

D. Within-domain models

Logistic regression (L2-regularised). **Random forest** [56]: bagged trees, $\hat{p}(x) = \frac{1}{B}\sum_b T_b(x)$. **Gradient boosting** [55] minimises $\sum_i \ell(y_i, \hat{y}_i) + \sum_m \Omega(f_m)$ with $\Omega(f) = \gamma T + \frac{1}{2}\lambda\|w\|^2$; with second-order Taylor expansion each leaf weight is $w_j^\star = -G_j/(H_j + \lambda)$ and the split gain is

$$\tfrac{1}{2}\Big[\tfrac{G_L^2}{H_L+\lambda} + \tfrac{G_R^2}{H_R+\lambda} - \tfrac{(G_L+G_R)^2}{H_L+H_R+\lambda}\Big] - \gamma. \tag{6}$$

Isolation forest [13]: anomaly score $s(x, n) = 2^{-\mathbb{E}[h(x)]/c(n)}$ with $c(n) = 2H(n-1) - 2(n-1)/n$, H the harmonic number. **Autoencoder**: trained on legitimate training points only, with score $\|x - \psi(\varphi(x))\|_2^2$. Following the evidence that tree ensembles remain the strongest tabular family at this scale [57], [58], gradient boosting is the primary within-domain comparator; an MLP/ResNet or FT-Transformer [59] and TabPFN for the small HI set [60] serve as secondary baselines.

E. Domain adaptation

After alignment, source and target samples live in $\mathcal{Z}$. Let $G_f : \mathcal{Z} \to \mathbb{R}^m$ be a feature extractor and $G_y : \mathbb{R}^m \to [0, 1]$ a classifier. All three methods minimise

$$\min_{G_f, G_y} \;\; \hat{R}_s^\omega(G_y \circ G_f) + \alpha\,\mathcal{L}_{\text{align}}(G_f) + \beta\,\hat{R}_{t,b}^{\omega_t}(G_y \circ G_f), \tag{7}$$

where $\hat{R}_s^\omega$ is the weighted source risk, $\hat{R}_{t,b}$ the risk on the b labelled target examples ($\beta = 0$ when $b = 0$), and $\mathcal{L}_{\text{align}}$ differs by method.

a) DANN.: Add a domain classifier $G_d : \mathbb{R}^m \to [0, 1]$ with domain labels $d = 0$ (source), $d = 1$ (target):

$$\begin{aligned} &E(\theta_f, \theta_y, \theta_d) = \hat{R}_s^\omega \\ &-\lambda\Big[\tfrac{1}{n_s}\sum_i \mathcal{L}_d\big(G_d(G_f(x_i^s)), 0\big) + \tfrac{1}{n_t}\sum_j \mathcal{L}_d\big(G_d(G_f(x_j^t)), 1\big)\Big] \end{aligned} \tag{8}$$

and seek the saddle point $(\hat{\theta}_f, \hat{\theta}_y) = \arg\min E(\cdot, \cdot, \hat{\theta}_d)$, $\hat{\theta}_d = \arg\max E(\hat{\theta}_f, \hat{\theta}_y, \cdot)$ [35]. The adaptation weight follows $\lambda_p = \frac{2}{1+e^{-\gamma p}} - 1$, $p \in [0, 1]$ training progress, $\gamma = 10$.

Proposition 7 (Gradient reversal realises the saddle point)**.** *Define a layer R_λ with $R_\lambda(z) = z$ and $\partial R_\lambda/\partial z = -\lambda I$. Minimising $\hat{R}_s^\omega + \mathcal{L}_d$ with the domain head fed through R_λ updates θ_d with $+\nabla_{\theta_d}\mathcal{L}_d$ (descent on $\mathcal{L}_d$) and θ_f with $\nabla_{\theta_f}\hat{R}_s^\omega - \lambda\nabla_{\theta_f}\mathcal{L}_d$, i.e. stochastic descent on E for (θ_f, θ_y) and ascent on E for θ_d (since $-\lambda\mathcal{L}_d$ enters E with negative sign, minimising $\mathcal{L}_d$ in θ_d maximises E).*

Proof. The chain rule through R_λ gives $\partial\mathcal{L}_d/\partial\theta_f = -\lambda\,(\partial\mathcal{L}_d/\partial z)(\partial z/\partial\theta_f)$ while $\partial\mathcal{L}_d/\partial\theta_d$ is unaffected. □

Remark 2. *The reversal coefficient must appear* once*: either in the layer (with unit weight on $\mathcal{L}_d$) or as an explicit loss weight with a plain identity layer, not both. Applying both makes the extractor see $-\lambda^2\nabla\mathcal{L}_d$ and the discriminator $\lambda\nabla\mathcal{L}_d$, silently changing the schedule of [35]; we apply it once.*

By (1), driving a good domain discriminator to chance reduces the proxy $\mathcal{A}$-distance $d_{\mathcal{A}} = 2(1-2\epsilon_d)$, where ϵ_d is the discriminator's error, which we report as a diagnostic. Conditional variants such as CDAN [37] would align joint structure but depend on target pseudo-predictions that are dominated by the majority class at a 0.5% prior, so we use the class-conditional criterion below instead.

b) Subdomain (class-conditional) MMD.: Let k be a Gaussian kernel $k_\sigma(u,v) = \exp(-\|u-v\|^2/2\sigma^2)$ with RKHS $\mathcal{H}_k$ and feature map ψ. The (biased) squared maximum mean discrepancy between samples $\{u_i\}_1^m$, $\{v_j\}_1^n$ is

$$\widehat{\mathrm{MMD}}^2 = \frac{1}{m^2}\sum_{i,i'}k(u_i,u_{i'}) + \frac{1}{n^2}\sum_{j,j'}k(v_j,v_{j'}) - \frac{2}{mn}\sum_{i,j}k(u_i,v_j). \tag{9}$$

Proposition 8 (MMD properties [39])**.** $\mathrm{MMD}^2(P,Q) = \|\mathbb{E}_P\psi(u) - \mathbb{E}_Q\psi(v)\|^2_{\mathcal{H}_k} = \sup_{\|f\|_{\mathcal{H}_k}\le 1}\big(\mathbb{E}_P f - \mathbb{E}_Q f\big)^2$*. For a characteristic kernel (Gaussian on $\mathbb{R}^d$),* $\mathrm{MMD}(P,Q) = 0 \iff P = Q$*; the unbiased estimator excluding diagonal terms has expectation* MMD^2.

Global MMD [38] matches marginals and therefore inherits Prop. 9. The *class-conditional* (subdomain) variant matches conditionals $p(z \mid y = c)$ [42]:

$$\mathcal{L}_{\mathrm{LMMD}} = \frac{1}{C}\sum_{c=1}^{C}\Big\|\sum_i w^s_{ic}\,\psi(G_f(x^s_i)) - \sum_j w^t_{jc}\,\psi(G_f(x^t_j))\Big\|^2_{\mathcal{H}_k}, \tag{10}$$

with $w^s_{ic} = y_{ic}/\sum_{i'} y_{i'c}$ (one-hot source labels) and $w^t_{jc} = \hat{y}_{jc}/\sum_{j'}\hat{y}_{j'c}$ using soft predictions; with only $C = 2$ classes and a rare positive class, the fraud weights are carried by few examples, so we use a multi-bandwidth kernel $\sum_u k_{\sigma_u}$, $\sigma_u \in \{0.5, 1, 2, 4\}$ (justified by Prop. 13) and report per-class sample counts. Because the conditional distributions are matched, the bound of Prop. 9 does not apply to $\mathcal{L}_{\mathrm{LMMD}}$ to the same extent, in the spirit of generalised label shift [47].

c) HEN-style attention-gated network.: A feature-attention gate $a(z) = \mathrm{softmax}\big(W_2\tanh(W_1 z + b_1) + b_2\big) \in \Delta^{d-1}$ re-weights inputs, $\tilde{z} = z \odot a(z)$, followed by an encoder g_ϕ and a linear head: $h(z) = \sigma(w^\top g_\phi(\tilde{z}) + b)$. Stage 1 minimises $\hat{R}^\omega_s$ ($\beta = 0$). Stage 2 fine-tunes on the b labelled target examples from the warm start θ_s with a smaller step and an L2-SP anchor [54]:

$$\hat{\theta}_t = \arg\min_\theta\ \hat{R}^{\omega_t}_{t,b}(\theta) + \tfrac{\eta}{2}\|\theta - \theta_s\|_2^2. \tag{11}$$

Remark 3. *Because the softmax sums to one, each gated coordinate is scaled by roughly $1/d$; batch normalisation after the first linear layer removes the resulting scale effect, but attention weights are then comparable only within, not across, models. Attention is a model-internal diagnostic, not a faithful explanation; explanations are obtained with SHAP (Section V-G). Since this network uses target labels, it competes in the label-budget setting $b > 0$ against a* target-only *model trained on the same b labels; Section VIII defines the transfer gain. We call it "HEN-style" because it borrows the attention idea of [6] but operates on tabular concept features, not behaviour sequences.*

Algorithm 1 Cross-sector transfer (one directed pair $s \to t$)

1: **Input:** $\mathcal{S}_s$, unlabelled $\{x^t_j\}$, budget b labelled target pairs $\mathcal{T}_b$, concept tables (G_{kc}, σ)
2: Split each sector (chronological/grouped); fit all preprocessors on train only
3: $z \leftarrow \phi_k(x)$ for $k \in \{s,t\}$ ▷ Eq. (4)
4: Estimate $\hat{\pi}_t$ (Prop. 11); set $\omega = \frac{\hat{\pi}_t/(1-\hat{\pi}_t)}{\pi_s/(1-\pi_s)}$
5: Train G_f, G_y by Eq. (7) with $\mathcal{L}_{\mathrm{align}} \in \{\mathrm{DANN}, \mathrm{LMMD}\}$; for the gated network pre-train then fine-tune (Eq. (11))
6: Select hyper-parameters by source/importance-weighted validation or CV on $\mathcal{T}_b$ (no target-test labels)
7: Calibrate $q \to p$ (Prop. 5); threshold at τ_q (Prop. 6); report $\Delta_m(b)$ vs. target-only

F. Stacked ensemble

Base learners $f_1, \ldots, f_M$ (LR, RF, XGB) produce out-of-fold predictions $z_{im} = f_m^{(-\kappa(i))}(x_i)$ where $\kappa(i)$ is the fold of i; a logistic meta-learner is fit on $z_i = (z_{i1}, \ldots, z_{iM})$ [61]. Resampling must be inside the fold loop so that OOF predictions do not see synthetic neighbours of the held-out point.

G. Explainability

For a model f and feature set F ($|F| = M$), the Shapley attribution of feature j at x is

$$\phi_j(x) = \sum_{S\subseteq F\setminus\{j\}}\frac{|S|!\,(M-|S|-1)!}{M!}\big[f_{S\cup\{j\}}(x) - f_S(x)\big], \tag{12}$$

the unique allocation satisfying efficiency ($\sum_j \phi_j = f(x) - \mathbb{E}f$), symmetry, dummy and additivity [63]; TreeSHAP computes it exactly for tree ensembles in $O(TLD^2)$ [64]. LIME fits $g = \arg\min_{g\in\mathcal{G}}\mathcal{L}(f,g,\pi_x) + \Omega(g)$ locally [65]. Both have limits that we respect: Shapley importances depend on the value function and on feature correlation [66], and perturbation-based explainers can be unreliable off the data manifold [67]. For cross-sector comparison, importances are aggregated to the three concepts, $I_c = \sum_{j\in G_{kc}}\overline{|\phi_j|}$ normalised to sum to one per sector, and compared by Kendall's τ with a permutation test; raw feature importances are *not* comparable across sectors because feature sets differ. Explanations are computed on held-out data only.

VI. Implementation Architecture and Algorithms

This section fixes the algorithmic structure that the corrected code base implements. Fig. 2 shows the end-to-end data flow

Algorithm 2 Health-insurance label construction (`labels_hi.py`)

Require: claims $\mathcal{C}$ with S, M, T and G-DRG; tariff table $\mathcal{A}$; tolerance $\tau = 1$

$A \leftarrow$ left-join of $\mathcal{C}$ with $\mathcal{A}$ on G-DRG (upper-cased, stripped); parse S, M, T numerically
for each claim **do**
  $r_1 \leftarrow \mathbf{1}[A \neq \varnothing \wedge S > A]$, $r_2 \leftarrow \mathbf{1}[A \neq \varnothing \wedge M > A/2]$, $r_3 \leftarrow \mathbf{1}[A \neq \varnothing \wedge T > A]$
  $r_4 \leftarrow \mathbf{1}[S, M, T \text{ all observed} \wedge |T - (S + M)| > \tau]$ ▷ not evaluable if M missing
  dq flags: missing S/M/T; observed non-positive; no tariff match
  $y \leftarrow \mathbf{1}[\sum_{i \le 4} r_i > 0]$; **if** $A = \varnothing \wedge y = 0$ **then** $y \leftarrow$ `NaN`
end for
return labels y, rule and dq flags, audit counts

Algorithm 3 Leakage-free preprocessing (`splits.py`, `preprocess.py`, `leakage.py`)

Require: frame X, labels y, domain k, grouping/time key

Drop identifier and rule-derived columns (registry); **assert** no remaining feature has single-feature AUC ≥ 0.98
Add deterministic time features (parse timestamp first)
$(\mathcal{I}_{\mathrm{tr}}, \mathcal{I}_{\mathrm{va}}, \mathcal{I}_{\mathrm{te}}) \leftarrow$ chronological split (CC, MM) or grouped stratified split (HI); **assert** disjoint
Partition columns on X_{tr} into numeric / low-cardinality / high-cardinality
Numeric: median-impute (+missing indicator) $\rightarrow$ winsorise at $q_{0.99}(X_{\mathrm{tr}}) \rightarrow$ robust-scale
Low-card.: one-hot with unseen$\rightarrow$zeros; **High-card.:** OOF target encoding
for fold (a, b) of K-fold on $\mathcal{I}_{\mathrm{tr}}$ **do**
  encode rows in b using statistics of rows in a only: $\frac{n_c \bar{y}_c + m\bar{y}}{n_c + m}$
end for
Fit full-train statistics for val/test encoding; unseen category $\mapsto \bar{y}$
return $A_{\mathrm{tr}}, A_{\mathrm{va}}, A_{\mathrm{te}}$, fitted transformer

Algorithm 4 DANN with prior-corrected weighting

Require: source (z^s, y^s), unlabelled target z^t, epochs E, $\gamma = 10$, weight ω (Prop. 5)

Initialise G_f, G_y, G_d
for $e = 1$ **to** E **do**
  $\lambda \leftarrow 2/(1 + e^{-\gamma e/E}) - 1$ ▷ set reversal strength in R_λ
  for each batch pair $(z^s, y^s), z^t$ **do**
    $f^s, f^t \leftarrow G_f(z^s), G_f(z^t)$
    $\mathcal{L}_y \leftarrow$ weighted BCE; $\mathcal{L}_d \leftarrow \frac{1}{2}[\mathrm{CE}(G_d(R_\lambda f^s), 0) + \mathrm{CE}(G_d(R_\lambda f^t), 1)]$
    update θ by Adam on $\mathcal{L}_y + \mathcal{L}_d$ ▷ λ applied once
  end for
  evaluate source-validation AP; keep best weights; stop if patience exhausted
end for
return $G_y \circ G_f$; optionally estimate $\hat{\pi}_t$ (Prop. 11) and correct posteriors (Prop. 10)

Algorithm 5 Class-conditional (subdomain) MMD training

Require: source (z^s, y^s), unlabelled target z^t, bandwidths $\{\sigma_u\}$, weight α

for each iteration **do**
  sample class-balanced source batch (ensures $\geq$ a few positives) and target batch
  $f^s, f^t \leftarrow G_f(z^s), G_f(z^t)$; $\hat{y}^t \leftarrow G_y(f^t)$ (soft; detached)
  $w^s_{ic} \leftarrow y_{ic} / \sum_{i'} y_{i'c}$; $w^t_{jc} \leftarrow \hat{y}_{jc} / \sum_{j'} \hat{y}_{j'c}$
  $K \leftarrow \sum_u k_{\sigma_u}$ over $[f^s; f^t]$ ▷ σ_u relative to median distance
  $\mathcal{L}_{\mathrm{LMMD}} \leftarrow \frac{1}{C} \sum_c \big(w_c^{s\top} K_{ss} w_c^s + w_c^{t\top} K_{tt} w_c^t - 2 w_c^{s\top} K_{st} w_c^t \big)$
  step on $\mathcal{L}_y + \alpha \, \mathcal{L}_{\mathrm{LMMD}}$
end for

with the module that realises each stage; Fig. 3 shows the information-flow rule that every stage obeys; Fig. 4 unrolls the adversarial training loop. Algorithms 2–7 give the procedures in executable detail, and Table V their cost.

The code realising Algorithms 2 and 3 (modules `labels_hi`, `leakage`, `splits`, `preprocess`) is covered by 18 unit tests on synthetic data that check, among others, that a missing drug charge is never labelled as fraud, that statistics are independent of test data, and that no group straddles a split; Algorithms 4–7 and the concept map are specified here and are the next implementation targets.

VII. Theoretical Analysis

This section collects results that determine what is achievable, and that dictate design choices in Section V. Several results instantiate known theorems in our setting and carry their citations (Props. 9–11, 8); the others (Prop. 4, Lemma 1, Props. 12, 13, 14, 15, 1) are derived here, several of them elementary, and their value lies in what they say about this application rather than in mathematical novelty.

A. Label shift limits invariant representations

Proposition 9 (Label shift limits invariant representations [3])**.** *Let* $\mathrm{JS}(\cdot, \cdot)$ *denote the Jensen–Shannon divergence with* $\log_2$ *[73] and* $d_{\mathrm{JS}} = \sqrt{\mathrm{JS}} \in [0, 1]$*. For any representation* $Z = G_f(X)$ *and classifier, the sum of source and target errors satisfies*

$$\varepsilon_s + \varepsilon_t \;\geq\; \tfrac{1}{2}\big(d_{\mathrm{JS}}(\mathcal{D}_s^Y, \mathcal{D}_t^Y) - d_{\mathrm{JS}}(\mathcal{D}_s^Z, \mathcal{D}_t^Z)\big)^2. \qquad (13)$$

In particular, perfect marginal alignment ($d_{\mathrm{JS}}(\mathcal{D}_s^Z, \mathcal{D}_t^Z) = 0$) *forces* $\lambda^\star$ *in* (1) *to be at least* $\frac{1}{2}\,\mathrm{JS}(\mathcal{D}_s^Y, \mathcal{D}_t^Y)$*.*

For Bernoulli priors the divergence has a closed form, which we use to evaluate the bound exactly.

Lemma 1 (Closed form for Bernoulli priors)**.** *With* $H_2(u) = -u \log_2 u - (1 - u) \log_2(1 - u)$*,* $\mathrm{JS}(\mathrm{Bern}(a), \mathrm{Bern}(b)) = H_2\big(\frac{a+b}{2}\big) - \frac{1}{2}\big(H_2(a) + H_2(b)\big)$.

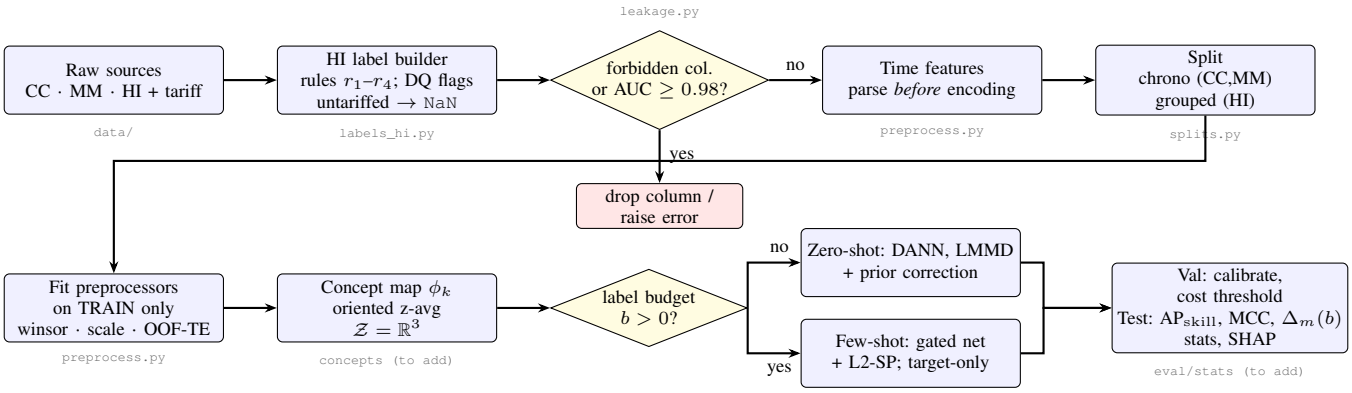


Fig. 2. End-to-end implementation flowchart with the code module for each stage. Every fitted component sees training rows only (Fig. 3); the leakage guard is a hard stop, not a warning.

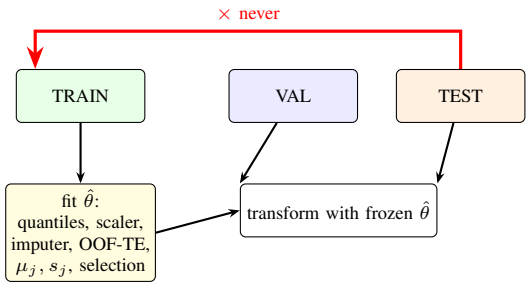


Fig. 3. Information-flow rule: statistics are learned from the training partition and frozen; validation and test are only transformed. The red path (test statistics informing preprocessing, encoding, selection or thresholds) is forbidden and covered by unit tests.

Algorithm 6 Gated network: pre-train and L2-SP fine-tune

Require: source data, b labelled target pairs $\mathcal{T}_b$, anchor weight η

1: Pre-train θ on weighted source risk until source-val AP plateaus; $\theta_s \leftarrow \theta$
2: Split $\mathcal{T}_b$ into inner folds (stratified); choose η, epochs by inner CV
3: Fine-tune: minimise $\hat{R}^{\omega_t}_{t,b}(\theta) + \frac{\eta}{2}\|\theta - \theta_s\|^2$ with small step
4: **return** $\hat{\theta}_t$; **baseline:** same architecture trained from scratch on $\mathcal{T}_b$ only

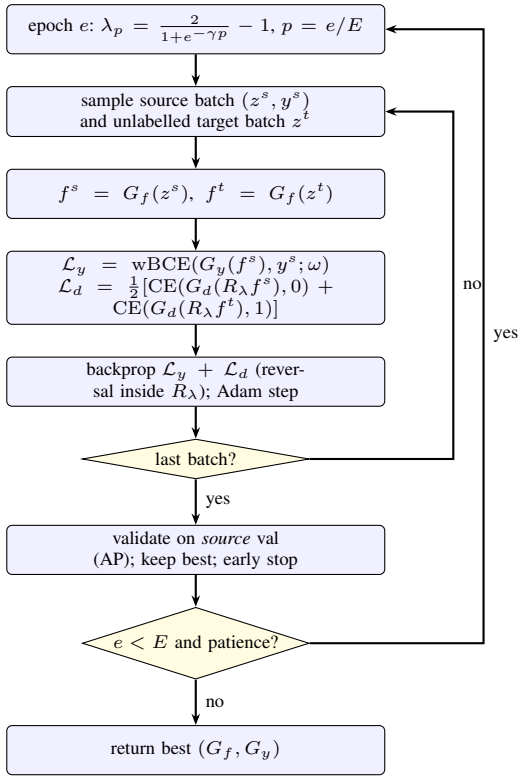


Fig. 4. Unrolled DANN training loop (Algorithm 4). The reversal coefficient is applied once, and model selection uses source validation, never target-test labels.

Proof. $\mathrm{JS}(P,Q) = H(M) - \frac{1}{2}H(P) - \frac{1}{2}H(Q)$ with $M = \frac{1}{2}(P+Q)$, and $M = \mathrm{Bern}(\frac{a+b}{2})$. □

Applying Lemma 1 to the priors of Table IV gives Table VI: Hence marginal alignment alone is expected to be harmful for transfers involving credit cards, and benign between mobile money and insurance. *Scope of the claim:* the theorem bounds the sum of unweighted 0–1 errors of a classifier acting on the aligned representation; it does not by itself bound AP, MCC or expected cost, and an error floor of 0.17 on the *sum* is not an impossibility result for every operating point. We therefore treat Table VI as a *directional prediction* to be tested with the metrics of interest (P1), not as a bound on those metrics. This is a testable prediction (Section IX). The remedy is to estimate target priors and correct the source (Props. 10, 11), to match conditionals [46], [47], or to use target labels, subject to the support condition that the map not destroy information [48].

B. Prior correction and estimation

Proposition 10 (Posterior correction under label shift [44])**.** *Assume label shift, $p_t(x \mid y) = p_s(x \mid y)$, with priors π_s, π_t. Then*

$$p_t(1 \mid x) = \frac{w_1\, p_s(1 \mid x)}{w_1\, p_s(1 \mid x) + w_0\, p_s(0 \mid x)}, \qquad w_y = \frac{\pi_t(y)}{\pi_s(y)}, \tag{14}$$

equivalently, posterior odds scale by $\frac{\pi_t/(1-\pi_t)}{\pi_s/(1-\pi_s)}$.

Algorithm 7 Transfer-gain evaluation with corrected statistics

Require: pairs (s,t), methods $\mathcal{M}$, budgets $\mathcal{B}$, seeds K

1: **for** each pair, budget b, seed $i = 1..K$ **do**
2: draw $\mathcal{T}_b$ from target-train; fit target-only baseline and every $m \in \mathcal{M}$ (Alg. 4–6)
3: calibrate on validation; threshold by Prop. 6; score on the fixed target test set
4: $d_i(m,b) \leftarrow M(\hat{h}_m) - M(\hat{h}_{\text{target-only}})$
5: **end for**
6: per (m,b): bootstrap CI of $\bar{d}$; corrected t (factor $\frac{1}{K} + \frac{n_{\text{te}}}{n_{\text{tr}}}$), Wilcoxon, Bayesian ROPE
7: apply Holm over all tests; report negative-transfer rate $= \Pr(d_i < 0)$

TABLE V
COMPUTATIONAL COST PER STAGE (n EXAMPLES, d INPUTS, HIDDEN WIDTH h, BATCH B, U BANDWIDTHS, E EPOCHS, T TREES, DEPTH D, L LEAVES).

Stage	Time
OOF target encoding	$O(n)$ per categorical column
Concept map	$O(n \sum_c \lvert G_c \rvert)$
DANN epoch	$O\big(n\,(dh + h^2)\big)$, two forwards per step
LMMD per step	$O(UB^2h)$ kernel + $O(B\,dh)$ network
Gated net (attention $d{\times}d$)	$O\big(n\,(d^2 + dh)\big)$ per epoch
TreeSHAP	$O(TLD^2)$ per instance
Stacking (K-fold, M base)	$K \cdot M$ base fits + meta fit

Proof. By Bayes, $p_t(1 \mid x) \propto \pi_t(1) p_t(x \mid 1) = \pi_t(1) p_s(x \mid 1) = \frac{\pi_t(1)}{\pi_s(1)}\, p_s(1 \mid x)\, p_s(x)$; normalise over $y \in \{0,1\}$. □

Proposition 11 (Black-box prior estimation [45])**.** *Let $\hat{y}$ be any fixed classifier with joint confusion matrix $C_{ij} = \mathbb{P}_s(\hat{y}{=}i, y{=}j)$ on the source and $\mu_i = \mathbb{P}_t(\hat{y}{=}i)$ on the target. Under label shift, $\mu = Cw$ with $w_j = \pi_t(j)/\pi_s(j)$; if C is invertible, $w = C^{-1}\mu$ and $\hat{\pi}_t(j) = w_j \pi_s(j)$.*

Proof. Under label shift $\mathbb{P}_t(\hat{y}{=}i \mid y{=}j) = \mathbb{P}_s(\hat{y}{=}i \mid y{=}j)$, hence $\mu_i = \sum_j \mathbb{P}_s(\hat{y}{=}i \mid y{=}j)\pi_t(j) = \sum_j \mathbb{P}_s(\hat{y}{=}i, y{=}j)\, \pi_t(j)/\pi_s(j)$. □

Label shift ($p(x \mid y)$ equal across sectors) is implausible between *different sectors*; we therefore apply Props. 10–11 *after* concept alignment as a recalibration step and test their effect (ablation A5), and we use a few labelled target points for prior estimation when $b > 0$.

C. Optimal use of scarce target labels

Combine source and target empirical errors as $\hat{\varepsilon}_\alpha(h) = \alpha\hat{\varepsilon}_t(h) + (1-\alpha)\hat{\varepsilon}_s(h)$, with $m = m_s + m_t$ examples in total and $\beta = m_t/m$. The combined-error bound of Ben-David et al. [33] has, schematically and with constants and logarithmic factors absorbed, the form

$$\varepsilon_t(\hat{h}_\alpha) \le \varepsilon_t(h_t^*) + C_1 \sqrt{\left(\tfrac{\alpha^2}{\beta} + \tfrac{(1-\alpha)^2}{1-\beta}\right) \tfrac{\mathrm{VC}\log m + \log(1/\delta)}{m}} + (1-\alpha)\,\Delta, \quad \Delta = \tfrac{1}{2}\hat{d}_{\mathcal{H}\Delta\mathcal{H}} + \lambda^\star + (\text{complexity}). \tag{15}$$

TABLE VI
LOWER BOUND ON $\varepsilon_s + \varepsilon_t$ UNDER PERFECT MARGINAL ALIGNMENT, $\frac{1}{2}\,\mathrm{JS}(\mathrm{Bern}(\pi_s), \mathrm{Bern}(\pi_t))$.

Pair	JS (bits)	d_{JS}	bound
CC↔MM	0.3310	0.575	0.166
CC↔HI	0.2295	0.479	0.115
MM↔HI	0.0127	0.113	0.006

We use the theorem only through the *functional form* of this bound; the proposition below is therefore a statement about the trade-off the bound encodes (how the optimal weight moves with Δ), not a numerical guarantee on target error.

Proposition 12 (Optimal source/target weighting, for bounds of the displayed form with $C_1 > 0$, $\Delta \ge 0$)**.** *Let $f(\alpha) = \alpha^2/\beta + (1-\alpha)^2/(1-\beta)$. (i) If $\Delta = 0$, the bound is minimised at $\alpha^* = \beta$ (each example weighted equally). (ii) If $\Delta > 0$, the minimiser satisfies $\alpha^* > \beta$: labelled target examples should be weighted* more *than their share of the sample, and increasingly so as Δ grows.*

Proof. $f'(\alpha) = 2\big(\frac{\alpha}{\beta} - \frac{1-\alpha}{1-\beta}\big) = \frac{2(\alpha-\beta)}{\beta(1-\beta)}$, so f is minimised at $\alpha = \beta$, proving (i). Let $g(\alpha) = C_1'\sqrt{f(\alpha)} + (1-\alpha)\Delta$ with $C_1' > 0$. Then $g'(\alpha) = C_1' \frac{f'(\alpha)}{2\sqrt{f(\alpha)}} - \Delta$; at $\alpha = \beta$, $f' = 0$ so $g'(\beta) = -\Delta < 0$, and g is decreasing at β; g is convex (sum of the convex $\sqrt{f}$, as the norm of an affine map, and a linear term), so its minimiser lies at $\alpha^* > \beta$. □

This justifies the larger weight on the b labelled target examples and the small step in Eq. (11); since Δ is large for cross-sector pairs, the optimum places most weight on target labels and transfer helps chiefly through the *initialisation*, which we test with the transfer gain $\Delta_m(b)$ of Section VIII.

D. MMD bandwidth and the information in the criterion

Proposition 13 (Bandwidth limits of the Gaussian MMD)**.** *For samples with finite second moments, as $\sigma \to \infty$,*

$$\mathrm{MMD}^2_\sigma(P,Q) = \frac{\|\mu_P - \mu_Q\|_2^2}{\sigma^2} + O(\sigma^{-4}), \tag{16}$$

i.e. a very wide kernel only matches means; as $\sigma \to 0$ with distinct support points the kernel matrix tends to the identity and the estimator carries no information about the distributions.

Proof. Expand $k_\sigma(u,v) = 1 - \frac{\|u-v\|^2}{2\sigma^2} + O(\sigma^{-4})$. Then $\mathrm{MMD}^2 = \mathbb{E}k(u,u') + \mathbb{E}k(v,v') - 2\mathbb{E}k(u,v) = \frac{1}{2\sigma^2}\big(2\mathbb{E}\|u-v\|^2 - \mathbb{E}\|u-u'\|^2 - \mathbb{E}\|v-v'\|^2\big)$. With $\mathbb{E}\|u-v\|^2 = \operatorname{tr}\Sigma_P + \operatorname{tr}\Sigma_Q + \|\mu_P - \mu_Q\|^2$ and $\mathbb{E}\|u-u'\|^2 = 2\operatorname{tr}\Sigma_P$ (similarly for Q), the covariance terms cancel and the bracket equals $2\|\mu_P - \mu_Q\|^2$. □

Consequently a single bandwidth either collapses to mean matching or becomes uninformative; a multi-bandwidth kernel $\sum_u k_{\sigma_u}$ [38] spans the regimes, and bandwidths are set relative to the median pairwise distance of the aligned features.

TABLE VII
FALSE-POSITIVE BUDGET FOR 50% PRECISION AT TPR = 0.8, AND PRECISION AT FPR = 1%, PER SECTOR PRIOR.

Sector	π	max FPR for $P^* = 0.5$	precision at FPR $= 1\%$
Credit card	0.0052	0.42%	0.295
Mobile money	0.5484	97.2%	0.990
Health ins.	0.4162	57.0%	0.983

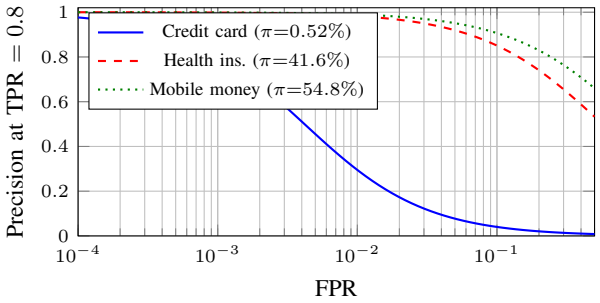


Fig. 5. Prop. 14: at the same detector quality (same TPR, FPR) precision differs by orders of magnitude across sectors, so F_1, precision and PR-AUC are not comparable between them.

E. Operating points and prevalence

Proposition 14 (Precision at a given operating point)**.** *For prior π, precision at true- and false-positive rates* TPR, FPR *is* $\mathrm{Prec} = \pi\,\mathrm{TPR}\,/(\pi\,\mathrm{TPR} + (1-\pi)\,\mathrm{FPR})$. *To reach precision P^* at a given* TPR*, the false-positive rate must satisfy*

$$\mathrm{FPR} \;\le\; \frac{\pi\;\mathrm{TPR}\,(1-P^*)}{(1-\pi)\,P^*}. \tag{17}$$

Proof. Bayes' rule on the confusion counts: $\mathrm{Prec} = \mathbb{P}(Y{=}1 \mid \hat{Y}{=}1) = \pi\,\mathrm{TPR}\,/\mathbb{P}(\hat{Y}{=}1)$ with $\mathbb{P}(\hat{Y}{=}1) = \pi\,\mathrm{TPR} + (1-\pi)\,\mathrm{FPR}$; solve for FPR. □

Corollary 1 (Prevalence-normalised comparison)**.** *The no-skill average precision equals π, so cross-sector comparison should use the skill score* $\mathrm{AP}_{\mathrm{skill}} = (\mathrm{AP} - \pi)/(1-\pi) \in [0,1]$, *alongside the expected cost of Prop. 6.*

Table VII and Fig. 5 make the practical point of precision–recall analysis [27] quantitative: a credit-card detector must hold $\mathrm{FPR} < 0.42\%$ merely to be right half the time at 80% recall, whereas the same detector quality in mobile money or insurance has a trivially lenient budget. This also explains why reporting ROC-AUC (insensitive to π) for cards, F_1 for mobile money and MCC for insurance makes cross-sector numbers incomparable.

F. Expected cost, ROC slope and calibration

Let $s(X)$ be a score with class-conditional densities f_1, f_0 for fraud and legitimate cases, and threshold t. With $\mathrm{TPR}(t) = \int_t^\infty f_1$ and $\mathrm{FPR}(t) = \int_t^\infty f_0$, the expected cost over n cases is

$$\mathrm{Cost}(t) = n\big[\pi\,c_{FN}\,(1-\mathrm{TPR}(t)) + (1-\pi)\,c_{FP}\;\mathrm{FPR}(t)\big]. \tag{18}$$

Proposition 15 (Iso-cost slope)**.** *An interior minimiser t^* of* Cost *satisfies* $\dfrac{f_1(t^*)}{f_0(t^*)} = \dfrac{(1-\pi)\,c_{FP}}{\pi\,c_{FN}}$*, i.e. the ROC curve at the optimum has slope* $\dfrac{d\,\mathrm{TPR}}{d\,\mathrm{FPR}} = \dfrac{(1-\pi)c_{FP}}{\pi c_{FN}}$*. For a calibrated score this is the posterior threshold $\tau^* = c_{FP}/(c_{FP}+c_{FN})$ of Prop. 6.*

Proof. $\mathrm{Cost}'(t) = n[\pi c_{FN} f_1(t) - (1-\pi)c_{FP} f_0(t)]$ since $\mathrm{TPR}' = -f_1$ and $\mathrm{FPR}' = -f_0$; setting it to zero gives the likelihood-ratio condition. The ROC slope is $\frac{d\,\mathrm{TPR}}{d\,\mathrm{FPR}} = \frac{-f_1}{-f_0}$. For calibrated $s = p$, Bayes gives $\frac{f_1}{f_0}(t) = \frac{1-\pi}{\pi}\frac{t}{1-t}$; equating with the slope above yields $t^*/(1-t^*) = c_{FP}/c_{FN}$, i.e. $t^* = c_{FP}/(c_{FP}+c_{FN})$. □

Prop. 15 shows why the same detector has different optimal operating points per sector even at identical costs: the slope scales with the odds $(1-\pi)/\pi$, which is 191 for cards against 0.82 for mobile money. Thresholds are chosen on validation data and the realised cost is reported with a bootstrap interval.

a) Calibration.: A score p is calibrated if $\mathbb{P}(Y{=}1 \mid p) = p$. We report the Brier score $\frac{1}{n}\sum_i (p_i - y_i)^2$ and the binned expected calibration error $\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{n}\,|\bar{y}_b - \bar{p}_b|$, where $\bar{y}_b, \bar{p}_b$ are the observed rate and mean score in bin b. Resampling and weighting shift scores by a known odds factor (Prop. 5), so calibration is assessed *after* applying the inverse correction; ranking metrics (AP, AUC) are unaffected by this monotone map, but thresholded and cost-based decisions are.

b) Alert-budget metric.: Investigation capacity, not a threshold, is usually the binding constraint. For an alert budget of the top κ fraction of cases we report $\mathrm{Prec@}\kappa = \frac{1}{\lceil \kappa n \rceil}\sum_{i \in \text{top-}\kappa} y_i$ and recall at κ; by Prop. 14 the achievable precision at a fixed κ is again bounded by the sector's prevalence, so κ is chosen per sector from operational capacity.

G. Resampled significance tests

For K resampled train/test splits with train size n_{tr} and test size n_{te}, the sample variance of the paired differences $\hat{\sigma}_d^2$ underestimates the variance of $\bar{d}$ because the estimates are positively correlated through overlapping training data. The corrected variance estimator is $(\frac{1}{K} + \frac{n_{\mathrm{te}}}{n_{\mathrm{tr}}})\hat{\sigma}_d^2$ [29]; with it the statistic $t = \bar{d}/\sqrt{(\frac{1}{K} + \frac{n_{\mathrm{te}}}{n_{\mathrm{tr}}})\hat{\sigma}_d^2}$ is compared to a Student t_{K-1}. For $K = 20$ and a 70/15/15 split ($n_{\mathrm{te}}/n_{\mathrm{tr}} = 0.214$) the factor is 0.264 versus 0.050 for the naive test, i.e. standard errors are about 2.3 times larger: significance claims made with the naive test on repeated splits are over-confident by that factor.

VIII. Experimental Protocol

a) Splits.: CC and MM: chronological 70/15/15 by transaction time (Sparkov's published train/test boundary is respected). HI: grouped stratified split by patient identifier.

b) Regimes.: (R1) within-domain; (R2) zero-shot transfer ($b = 0$); (R3) few-shot transfer with $b \in \{10, 50, 100, 500\}$ labels, drawn with 20 random seeds; (R4) *target-only* baseline using the same b labels. The *transfer gain* of method m at budget b is

$$\Delta_m(b) = M\big(\hat{h}_m^{(b)}\big) - M\big(\hat{h}_{\text{target-only}}^{(b)}\big), \tag{19}$$

for a metric M; positive Δ indicates real transfer rather than label supervision alone, and negative Δ is reported as *negative transfer* [51].

c) Model selection and thresholds.: No target-test labels, and for $b = 0$ no target labels at all, are used for early stopping, checkpoint choice or thresholds. Hyper-parameters are selected by (a) source validation, (b) importance-weighted validation [53] with prior-corrected weights (since density-ratio weights are heavy-tailed under extreme prior mismatch), or (c) for $b > 0$, cross-validation within the b labelled target examples. Thresholds follow Prop. 6.

d) Metrics.: With confusion counts TP, FP, TN, FN (MCC [28]; precision–recall analysis [27]):

$$F_1 = \frac{2TP}{2TP + FP + FN}, \tag{20}$$

$$\mathrm{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}, \tag{21}$$

$$\mathrm{AUC} = \mathbb{P}\big(s(X^+) > s(X^-)\big) + \tfrac{1}{2}\mathbb{P}\big(s(X^+) = s(X^-)\big), \tag{22}$$

$$\mathrm{AP} = \sum_n (R_n - R_{n-1}) P_n. \tag{23}$$

Average precision (the step-wise sum) is reported instead of the trapezoidal area $\mathrm{auc}(R, P)$, which interpolates linearly between operating points and is optimistic. Metrics are reported in *all* sectors (not a different metric per sector), as $\mathrm{AP}_{\mathrm{skill}}$ alongside MCC, expected cost, calibration error, and precision at a fixed alert budget.

e) Statistics.: Over K seeds the paired difference $d_i = M_i^A - M_i^B$ is tested with the corrected resampled t-statistic (Section VII), with Wilcoxon signed-rank as a non-parametric check, a Bayesian correlated t-test with a region of practical equivalence [31], DeLong's test for correlated AUCs [74], bootstrap percentile intervals ($B = 1000$), and Holm's step-down correction [32]: sort $p_{(1)} \le \cdots \le p_{(m)}$ and reject $H_{(i)}$ while $p_{(i)} \le \alpha/(m - i + 1)$. Comparisons are never pooled across heterogeneous transfer scenarios: a paired test over six scenario scores treats unlike units as exchangeable, and rank-based multi-dataset tests [30] have almost no power with three datasets, so cross-sector summaries are descriptive.

f) Directed scenarios and predictions.: Table VIII lists the six directed pairs with the label-shift floor of Table VI and the qualitative prediction each is meant to test.

TABLE VIII
TRANSFER SCENARIOS, LABEL-SHIFT FLOOR $\frac{1}{2}$ JS, AND WHAT EACH TESTS.

Pair $s \to t$	Floor	Prediction under test
CC→MM	0.166	marginal DA harmful; prior correction needed
CC→HI	0.115	marginal DA harmful; HI amount concept empty
MM→CC	0.166	rare target positives; prior estimation fragile
MM→HI	0.006	DA benign; best chance of real transfer
HI→CC	0.115	tiny source (n=2655) to huge target: weak
HI→MM	0.006	DA benign; source-limited

TABLE IX
WITHIN-DOMAIN RESULTS (R1), TEST SET, MEAN ± 95% BOOTSTRAP CI OVER SEEDS.

Sector	Model	$\mathrm{AP}_{\mathrm{skill}}$	MCC	Cost
CC	LR / RF / XGB / Stack	**TBD**	**TBD**	**TBD**
MM	LR / RF / XGB / Stack	**TBD**	**TBD**	**TBD**
HI	LR / RF / XGB / Stack / TabPFN	**TBD**	**TBD**	**TBD**

TABLE X
TRANSFER GAIN $\Delta_m(b)$ IN $\mathrm{AP}_{\mathrm{skill}}$ OVER THE TARGET-ONLY BASELINE; ROWS ARE DIRECTED PAIRS, COLUMNS LABEL BUDGETS.

Pair	Method	b=0	10	100	500
CC→MM	DANN	**TBD**	**TBD**	**TBD**	**TBD**
	LMMD	**TBD**	**TBD**	**TBD**	**TBD**
	Gated+L2-SP	–	**TBD**	**TBD**	**TBD**
… (all six directed pairs)					

g) Ablations and baselines.: (A1) remove alignment (raw positional features; a negative control whose result depends on column order); (A2) alignment without orientation σ_j; (A3) each adaptation loss removed; (A4) global vs. class-conditional MMD; (A5) label-shift-corrected source weights; (A6) CORAL [40]; (A7) gradient boosting with a domain indicator and, as strong tabular references, XGBoost [55], FT-Transformer [59] and TabPFN [60]; (A8) resampling vs. reweighting; (A9) HI without rule inputs and with a rule-only baseline, to measure circularity (Prop. 2). Cross-sector tabular shift benchmarks [52] motivate reporting no-transfer and gradient-boosting baselines in every table.

IX. RESULTS

All numbers in this section are placeholders and will be generated by the leakage-free protocol of Section VIII; no result is reported until the full pipeline (preprocessing fitted on training data only, grouped and chronological splits, target-label-free model selection) has been re-executed end to end.

Predictions to be confirmed or refuted. (P1) DANN and global MMD underperform the target-only baseline for CC→MM and CC→HI, where Table VI predicts an irreducible error floor, and are neutral for MM↔HI. (P2) Class-conditional LMMD and label-shift-corrected weights narrow this gap. (P3) Gains of the gated network vanish once compared with the target-only baseline at equal b unless the source representation is genuinely informative (Prop. 12). (P4) Concept-level SHAP rankings agree more strongly between MM and HI than with CC. (P5) Full-feature within-domain models weakly dominate concept-space models (Prop. 4). Each prediction is falsifiable, and refutations will be reported as negative transfer.

X. DISCUSSION AND THREATS TO VALIDITY

Synthetic data. Two sectors are simulated; conclusions about relative method ranking may not transfer to production traffic, and simulators encode the generator's own fraud heuristics (e.g. balance-consistency features are close to label-defining in PaySim-type generators) [14], [18], [19]. The

54.8% fraud rate of the MoMTSim extract is implausible for real mobile money and its generating scenario should be documented. **Label validity (HI).** Rule labels measure tariff non-compliance, not adjudicated fraud, and are circular with billing features (Prop. 2); the missing-drug-charge artefact (Prop. 1) must be removed or reported separately. **Small n.** Health insurance has $n = 2655$; confidence intervals are wide and few-shot results are high-variance. **Concept map.** Group membership G_{kc} and orientations σ_j are expert-specified; sensitivity to these choices is an explicit ablation (A2). **Explanation validity.** Attributions on aligned features inherit the limits in [66], [67]. **Fairness and regulation.** Flagging rates should be audited across region and facility type [68], protected attributes may exist in only one schema, and explanations may be required for automated decisions [62]; thresholds should reflect investigation capacity. **Claims of priority.** The novelty statements in Section II-H rest on the searches performed and should be re-verified in Scopus, IEEE Xplore and the ACM Digital Library.

A. Deployment, monitoring and governance

Cross-sector transfer is most valuable at *cold start*, when a new product has few labels; deployment must therefore plan for the label arrival that follows. (i) *Drift monitoring.* For each concept coordinate and for the score, the population stability index $\mathrm{PSI} = \sum_{i=1}^{B}(p_i - q_i)\ln(p_i/q_i)$ compares the binned reference distribution p with the live distribution q; a persistent rise triggers recalibration. (ii) *Prior tracking.* Because the posterior odds scale linearly with the prior ratio (Prop. 10), the live fraud rate estimated by the black-box estimator (Prop. 11) can be used to rescale the threshold without retraining, as long as $p(x \mid y)$ is stable. (iii) *Label-delay awareness.* Feedback arrives on alerts first and on the rest later [9]; the target-only baseline of Section VIII should be re-fitted as labels accumulate, and the transfer gain $\Delta_m(b)$ read as a function of b shows the label count at which the transferred model stops being useful. (iv) *Governance.* Decisions that significantly affect individuals may require explanation and human review [62]; explanation logs should carry the concept-level attribution of Section V-G, and flag rates should be audited across region, facility type and age [68].

B. Data and code availability

The raw Sparkov and MoMTSim files are public, and the NHIA claims are available only under a data-use agreement; the repository therefore ships code, label audit scripts and manifests, not data. The corrected pipeline, its unit tests and the paper source are versioned in the project repository, and all reported numbers will be regenerated from raw data by one command before submission.

C. Ethics and data protection

The health-insurance claims contain patient-level fields. Use of the data requires the data-use agreement of the supplying authority, removal of direct identifiers (folder and insurance identifiers are used only to form patient groups for splitting, then discarded), and, where required, institutional ethical approval; the statements for approval, consent waiver and storage are to be completed by the authors (VERIFY). Flag decisions should not be used as the sole basis for denying a claim.

D. Limitations and future work

Three extensions follow naturally. First, *learned* concept alignment (per-sector encoders trained jointly with a concept-consistency loss) could replace expert groups G_{kc}, at the price of the identifiability issues discussed in [48]. Second, *unbalanced optimal transport* would relax the mass-preservation assumption that makes classical transport brittle under prior shift [41]. Third, *real* mobile-money and card logs, when accessible under agreement, would test whether conclusions drawn from simulators [2], [14] survive contact with production traffic.

XI. Conclusion

We gave a complete mathematical specification of a heterogeneous cross-sector fraud-transfer pipeline: an affine-invariant semantic alignment with a Bayes-risk accounting, a unified view of adversarial, kernel and fine-tuning adaptation, a closed-form label-shift analysis showing where invariance methods must fail, corrections and estimators that restore consistency, optimal target-label weighting, prior-aware decision theory with sector-specific operating budgets, and an evaluation protocol free of the leakage channels that most threaten such studies. The label-shift analysis predicts concrete failure modes of invariant-representation methods for Ghana's sector pairs, which the corrected experiments will test. If the predictions hold, the practical lesson is that cross-sector fraud transfer should be built on prior correction and target labels, not on distribution matching alone; if they fail, the negative result is itself informative.

Appendix A
Concept Tables and Open Design Issue

Table XI gives the provisional assignment G_{kc} and orientation σ_j (all $+1$ unless noted). Orientations are expert hypotheses to be validated by the sign of the univariate association with fraud on the training partition; ablation A2 measures their effect. Identifier-like fields are excluded in every sector.

TABLE XI
Provisional concept groups (features after leakage removal).

Sector	Amount anomaly	Counterparty risk	Structural distance
CC	amount, robust amount z, large-amount flag	merchant and category (out-of-fold TE)	card–merchant distance, unusual hour, city-size flag
MM	amount, robust amount z	destination zero-balance flag, agent proxy	balance-consistency error (orig., dest.), unusual hour
HI	*none admissible* (see below)	DRG volume, specialty	length of stay, month/weekday of attendance

a) Open issue (health insurance).: The natural "amount anomaly" of a claim is its deviation from the approved tariff, which is exactly an input of the labelling rules (Prop. 2); any such feature makes the HI task circular. We therefore set the HI amount concept to 0 (empty group) in the main experiments and report an auxiliary analysis with a non-circular proxy (e.g. procedure-complexity-adjusted charge) and with labels from a source other than amount rules once available. This makes the HI concept space effectively two-dimensional, which the transfer-gain results must be read against.

APPENDIX B
LABEL DECISION FLOW

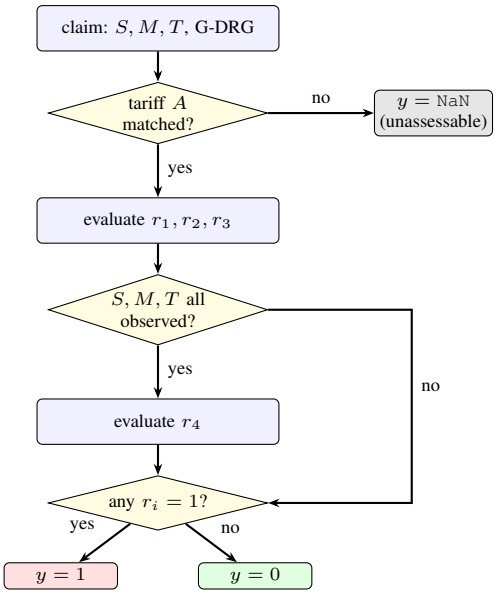


Fig. 6. Decision flow of Algorithm 2. Unmatched claims with no firing rule are excluded rather than labelled legitimate; missing M disables r_4 instead of producing a fraud label. (For unmatched claims, r_4 is still evaluated in the implementation; the figure shows the main path.)

APPENDIX C
CONFIGURATION AND REPRODUCIBILITY

TABLE XII
EXPERIMENTAL CONFIGURATION (VALUES TAKEN FROM THE CURRENT CODE UNLESS MARKED **TBD**).

Item	Setting
Encoder (DANN)	MLP $d \to 128 \to 64$, BatchNorm, ReLU, dropout 0.3
Heads	label: $64 \to 64 \to 1$; domain: $64 \to 64 \to 2$
Optimiser	Adam, lr 10^{-3}, weight decay 10^{-4}, cosine schedule
Epochs E, γ	50, 10
Fine-tuning	lr 3×10^{-4}, L2-SP $\eta =$ **TBD**(inner CV)
MMD bandwidths	$\{0.5, 1, 2, 4\}\times$ median pairwise distance
Winsorising / TE	$q = 0.99$; smoothing $m = 20$, 5-fold OOF
Splits	70/15/15; chronological (CC, MM), grouped (HI)
Seeds / budgets	$K = 20$; $b \in \{10, 50, 100, 500\}$
Bootstrap, α	$B = 1000$, 0.05 with Holm
RNG	fixed seeds for `numpy`, `torch` and all sampling

All intermediate frames, manifests (sizes, prevalence, dropped columns, label audit) and per-seed result tables are written to disk and versioned with the code; every reported number is regenerated by a single command from raw data.

APPENDIX D
PROOF DETAILS

Proof of Prop. 8. By the reproducing property, $\mathbb{E}_P f = \langle f, \mu_P \rangle_{\mathcal{H}_k}$ with $\mu_P = \mathbb{E}_P \psi(u)$. Hence $\sup_{\|f\| \le 1} (\mathbb{E}_P f - \mathbb{E}_Q f) = \|\mu_P - \mu_Q\|_{\mathcal{H}_k}$, and squaring gives the identity. Expanding the norm, $\|\mu_P - \mu_Q\|^2 = \mathbb{E}k(u, u') + \mathbb{E}k(v, v') - 2\mathbb{E}k(u, v)$ with $u, u' \sim P$ and $v, v' \sim Q$ independent. For the unbiased estimator drop the $i = i'$ and $j = j'$ terms and normalise by $m(m-1)$ and $n(n-1)$; each retained term has expectation $\mathbb{E}k(u, u')$ or $\mathbb{E}k(v, v')$, so the estimator is unbiased. Injectivity of $P \mapsto \mu_P$ for characteristic kernels (Gaussian on $\mathbb{R}^d$) gives $\mathrm{MMD} = 0 \iff P = Q$ [39].

Proof of Prop. 9 (consequence for $\lambda^\star$). The sum bound is the cited theorem [3]. Because $\lambda^\star = \min_{h'} [\varepsilon_s(h') + \varepsilon_t(h')]$ is the smallest value that sum can take over $\mathcal{H}$, the bound with $d_{\mathrm{JS}}(\mathcal{D}_s^Z, \mathcal{D}_t^Z) = 0$ gives $\lambda^\star \ge \frac{1}{2} d_{\mathrm{JS}}(\mathcal{D}_s^Y, \mathcal{D}_t^Y)^2 = \frac{1}{2}\,\mathrm{JS}(\mathcal{D}_s^Y, \mathcal{D}_t^Y)$.

Convexity in Prop. 12. $\sqrt{f(\alpha)} = \|(\alpha/\sqrt{\beta}, (1-\alpha)/\sqrt{1-\beta})\|_2$ is the Euclidean norm of an affine function of α, hence convex; adding the linear term $(1-\alpha)\Delta$ preserves convexity. Since $g'(\beta) = -\Delta < 0$ and g is convex, g' is non-decreasing, so the unique zero of g' lies to the right of β.

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